

Direct Kinematic Model

	θ_i	d_i	a_i	α_i
1	θ_1	0	a_1	0
2	θ_2	0	a_2	0
3	θ_3	0	a_3	0

$$T_1^0 = \begin{bmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & a_1 \cos(\theta_1) \\ \sin(\theta_1) & \cos(\theta_1) & 0 & a_1 \sin(\theta_1) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_2^1 = \begin{bmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & a_2 \cos(\theta_2) \\ \sin(\theta_2) & \cos(\theta_2) & 0 & a_2 \sin(\theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^2 = \begin{bmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & a_3 \cos(\theta_3) \\ \sin(\theta_3) & \cos(\theta_3) & 0 & a_3 \sin(\theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = T_1^0 * T_2^1 * T_3^2$$

$$T_2^0 = T_1^0 * T_2^1 = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & a_1 \cos(\theta_1) + a_2 \cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & a_1 \sin(\theta_1) + a_2 \sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$T_3^0 = T_2^0 * T_3^2 = \begin{bmatrix} \cos(\theta_1 + \theta_2 + \theta_3) & -\sin(\theta_1 + \theta_2 + \theta_3) & 0 & a_1 \cos(\theta_1) + a_2 \cos(\theta_1 + \theta_2) + a_3 \cos(\theta_1 + \theta_2 + \theta_3) \\ \sin(\theta_1 + \theta_2 + \theta_3) & \cos(\theta_1 + \theta_2 + \theta_3) & 0 & a_1 \sin(\theta_1) + a_2 \sin(\theta_1 + \theta_2) + a_3 \sin(\theta_1 + \theta_2 + \theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$x = a_1 \cos(\theta_1) + a_2 \cos(\theta_1 + \theta_2) + a_3 \cos(\theta_1 + \theta_2 + \theta_3)$$

$$y = a_1 \sin(\theta_1) + a_2 \sin(\theta_1 + \theta_2) + a_3 \sin(\theta_1 + \theta_2 + \theta_3)$$

Inverse kinematics

$$x_w = x - L_3 \cos \phi \therefore x_w = L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)$$

$$y_w = y - L_3 \sin \phi \therefore y_w = L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)$$

Find θ_2

$$x_w^2 + y_w^2 = [L_1 \cos \theta_1 + L_2 \cos(\theta_1 + \theta_2)]^2 + [L_1 \sin \theta_1 + L_2 \sin(\theta_1 + \theta_2)]^2$$

$$x_w^2 + y_w^2 = [L_1^2 \cos^2 \theta_1 + 2L_1 L_2 \cos \theta_1 \cos(\theta_1 + \theta_2) + L_2^2 \cos^2(\theta_1 + \theta_2)] + [L_1^2 \sin^2 \theta_1 + 2L_1 L_2 \sin \theta_1 \sin(\theta_1 + \theta_2) + L_2^2 \sin^2(\theta_1 + \theta_2)]$$

$$x_w^2 + y_w^2 = L_1^2(\cos^2 \theta_1 + \sin^2 \theta_1) + L_2^2(\cos^2(\theta_1 + \theta_2) + \sin^2(\theta_1 + \theta_2)) + 2L_1 L_2(\cos \theta_1 \cos(\theta_1 + \theta_2) + \sin \theta_1 \sin(\theta_1 + \theta_2))$$

$$x_w^2 + y_w^2 = L_1^2 + L_2^2 + 2L_1 L_2 \cos \theta_2$$

$$\cos \theta_2 = \frac{x_w^2 + y_w^2 - L_1^2 - L_2^2}{2L_1 L_2} \rightarrow \cos \theta_2 = A$$

$$\sin^2 \theta + \cos^2 \theta = 1 \therefore \sin^2 \theta + A^2 = 1$$

$$\sin \theta = \pm \sqrt{1 - A^2}$$

$$\theta_2 = \text{atan2}(\pm \sqrt{1 - A^2}, A) \therefore \begin{cases} \theta_{2A} = \text{atan2}(\sqrt{1 - A^2}, A) \\ \theta_{2B} = \text{atan2}(-\sqrt{1 - A^2}, A) \end{cases}$$

